Department of Information Systems

Bachelor of Information Systems

Lab 2

IS301 Structured Programming

Weighting: 4%
ULO addressed: iii, iv, vi
Due date: @11:59PM, Saturday 22nd March 2025
Total Marks: 22

Selection Logic, Repetition Logic and Functions

Instructions:

1. Read the following problem carefully.
2. Develop an algorithm to solve this problem
3. You full name, student ID, date and the algorithm should be at the beginning of the program area.
4. Completed program should be submitted before the due date via IS301 page on Moodle.

Problem:

Write a simple guessing program in C. The computer is to generate a random number between 1 and 20. the user is given five tries to guess the right number. After each guess, you are to tell the user if the guessed number is greater than, less than or equal to the random number. If it is equal, no more guesses should be made, if the user hasn't guessed the number, **after five tries**, display the number with a message that the user should know it by now and terminate the program. Your design for this program should include a separate function to get the users guesses, a function to print the unsuccessful message, one to print the successful message and one to print the sorry message. Include your algorithm to solve this problem.

A possible successful dialog:

```
I am thinking of a number between 1 and 20.
Can you guess what it
is?                                          10
Your guess is low. Try
again:                                       15
Your guess is low. Try
again:                                       17
Your guess is high. Try
again:                                       16
Congratulations! You did it.
```

A possible unsuccessful dialogs:

```
I am thinking of a number between 1 and 20.
Can you guess what it is? 10
Your guess is low. Try
again:                                       20
Your guess is high. Try
again:                                       10
Your guess is low. Try
again:                                       18
Your guess is high. Try
again:                                       12
Sorry. The number was 15.
You should have gotten it by now.
Better luck next time.
```

Marking Criteria

Criteria	Description	Marks
Program Structure & Readability	- Code is well-structured and readable. - Proper indentation and spacing. - Meaningful variable and function names.	2
Random Number Generation	- Generates a random number between 1 and 20 correctly using rand().	2
User Input Handling	- Accepts user input correctly for guesses	2
Correct Use of Functions	- Implements separate functions for: o getting input, printing success/failure messages, o checking results. o Printing Hints	3
Implement of Selection Logic (if, if…else) statements	- Correctly determines whether the guess is too high, too low, or correct.	3
Implement Repetition Logic (Looping)	- Uses a **for loop** to handle **five attempts** correctly. Terminates if the correct guess is made.	3
User-Friendly Input & Output	- Provides clear prompts for user input. Outputs helpful hints for incorrect guesses.	2
Program Execution & Termination	- - Terminates immediately after a correct guess. - Displays correct message if all attempts are used.	3
Compilation & Error-Free Execution	- Code compiles without errors. Runs successfully with correct output.	2
	Total	**22**